United States Patent Application Publication	20250285928
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Lee; Sanghoon

SEMICONDUCTOR PACKAGE AND METHOD OF MANUFACTURING THE SAME

Abstract

A semiconductor package includes a first die structure including a support substrate having a first surface and a second surface opposite to the first surface and having an opening that extends from the first surface to the second surface, a first semiconductor chip disposed in the opening, and a gap filling layer that fills a gap between a sidewall of the opening and the first semiconductor chip; and a second die structure stacked on the first die structure, the second die structure including a second semiconductor chip that is electrically connected to the first semiconductor chip. An outer side surface of the support substrate and an outer side surface of the second semiconductor chip are positioned on the same plane.

Inventors: **Lee; Sanghoon (Suwon-si, KR)**

Applicant: **Samsung Electronics Co., Ltd.** (Suwon-si, KR)

Family ID: **1000008447299**

Appl. No.: **19/047681**

Filed: **February 07, 2025**

Foreign Application Priority Data

KR	10-2024-0033678	Mar. 11, 2024

Publication Classification

Int. Cl.: **H01L23/31** (20060101); **H01L21/56** (20060101); **H01L23/00** (20060101); **H01L23/14** (20060101); **H01L23/29** (20060101); **H01L25/07** (20060101); **H10B80/00** (20230101)

U.S. Cl.:

CPC **H01L23/3135** (20130101); **H01L21/56** (20130101); **H01L24/08** (20130101); **H01L24/09** (20130101); **H01L25/071** (20130101); H01L23/147 (20130101); H01L23/291 (20130101); H01L2224/08145 (20130101); H01L2224/09181 (20130101); H01L2225/06517 (20130101); H01L2225/06541 (20130101); H10B80/00 (20230201)

Background/Summary

PRIORITY STATEMENT
[0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0033678, filed on Mar. 11, 2024 in the Korean Intellectual Property Office (KIPO), the contents of which are herein incorporated by reference in their entirety.
BACKGROUND
1. Field
[0002] The invention relates to a semiconductor package and a method of manufacturing the semiconductor package. More particularly, the invention relates to a semiconductor package including a plurality of stacked semiconductor chips and a method of manufacturing the same.
2. Description of the Related Art
[0003] In order to manufacture a semiconductor package including stacked semiconductor chips, a plurality of first semiconductor chips may be placed on a carrier substrate and a gap filling layer may be formed to fill spaces between the plurality of first semiconductor chips to form a reconstructed wafer, and then, the reconstructed wafer may be attached to a target wafer including second semiconductor chips provided therein by a wafer-to-wafer bonding process. However, in the reconstructed wafer, a step difference may occur between an upper surface of the first semiconductor chip and an upper surface of the gap filling layer, thereby generating voids during the wafer-to-wafer bonding process. For example, a bevel step difference may occur in an edge region of the reconstructed wafer, thereby causing voids in the bevel region during the wafer-to-wafer bonding process.
SUMMARY
[0004] Aspects of the invention provide a semiconductor package having improved bonding quality. Aspects of the invention provide a method of manufacturing the semiconductor package.
[0005] According to example embodiments, a semiconductor package includes a first die structure including a support substrate having an opening therein, a first semiconductor chip disposed in the opening, and a gap filling layer that fills a gap between a sidewall of the opening and the first semiconductor chip. The semiconductor package further includes a second die structure stacked on the first die structure. The second die structure includes a second semiconductor chip that is electrically connected to the first semiconductor chip. The first semiconductor chip includes a first circuit substrate, a first front insulating layer formed on a first surface of the first circuit substrate and provided with first bonding pads, and a first backside insulating layer formed on a second surface opposite to the first surface of the first circuit substrate and provided with second bonding pads. The second semiconductor chip includes a second circuit substrate and a second front insulating layer formed on a first surface of the second circuit substrate and provided with third bonding pads. The second bonding pads are directly bonded to the third bonding pads respectively.
[0006] According to example embodiments, a semiconductor package includes a first die structure and a second die structure stacked on the first die structure. The first die structure includes a support substrate having a first surface and a second surface opposite to the first surface and having an opening that extends from the first surface to the second surface, a first semiconductor chip disposed in the opening, and a gap filling layer that fills a gap between a sidewall of the opening and the first semiconductor chip. The second die structure includes a second semiconductor chip

that is electrically connected to the first semiconductor chip. An outer side surface of the support substrate and an outer side surface of the second semiconductor chip may be positioned on the same plane.

[0007] According to example embodiments, a semiconductor package includes a first semiconductor chip, a plurality of die structures sequentially stacked on the first semiconductor chip, and a second semiconductor chip stacked on an uppermost die structure of the plurality of die structures. Each of the plurality of die structures includes a support substrate having an opening therein, a third semiconductor chip disposed in the opening, and a gap filling layer that fills a gap between a sidewall of the opening and the third semiconductor chip. An outer side surface of the support substrate of the uppermost die structure and an outer side surface of the second semiconductor chip may be positioned on the same plane.

[0008] According to example embodiments, in a method of manufacturing a semiconductor package, a support substrate having a first surface and a second surface opposite the first surface is provided. A recess is formed in the support substrate to have a predetermined depth from the first surface. A first semiconductor chip is disposed within the recess. A gap filling layer is formed to fill a gap between an outer side surface of the first semiconductor chip and a sidewall of the recess. A portion of the support substrate is partially removed to expose the first semiconductor chip. The removed portion of the support substrate is adjacent to the second surface of the support substrate. A second semiconductor chip is bonded on the support substrate to be electrically connected to the first semiconductor chip.

[0009] In example embodiments, the disposing of the first semiconductor chip in the recess may include: providing the first semiconductor chip including: a first circuit substrate, a plurality of through electrodes penetrating the first circuit substrate, and a first front insulating layer formed on a first surface of the first circuit substrate and provided with first bonding pads electrically connected to the plurality of through electrodes, and disposing the first semiconductor chip in the recess such that the first front insulating layer faces the support substrate.

[0010] In example embodiments, the method may further include: forming a first backside insulating layer on a second surface of the first circuit substrate provided with second bonding pads electrically connected to the plurality of through electrodes, and the second surface of the first circuit substrate is opposite to the first surface of the first circuit substrate; and removing a portion of the first circuit substrate to expose end portions of the plurality of through electrodes, and the removed portion of the first circuit substrate being adjacent to the second surface of the first circuit substrate. The forming of the gap filling layer may include: forming the gap filling layer on the first surface of the support substrate to cover the first semiconductor chip, and removing an upper portion of the gap filling layer to expose the end portions of the plurality of through electrodes.

[0011] In example embodiments, the bonding of the second semiconductor chip on the support substrate may include: providing the second semiconductor chip including a second circuit substrate and a second front insulating layer formed on a first surface of the second circuit substrate and provided with third bonding pads; and bonding the second front insulating layer of the second semiconductor chip to the first backside insulating layer of the first semiconductor chip.

[0012] In example embodiments, the bonding of the second front insulating layer to the first backside insulating layer may include directly bonding the third bonding pads to the second bonding pads.

[0013] In example embodiments, the method may further include forming conductive bumps respectively on the first bonding pads of the first semiconductor chip exposed by the first front insulating layer.

[0014] In example embodiments, the gap filling layers may be formed of silicon oxide.

[0015] According to example embodiments, a semiconductor package may include a first die structure and a second die structure stacked on the first die structure. The first die structure may include a support substrate having a recess therein, a first semiconductor chip disposed in the

recess, and a gap filling layer that fills a gap between a sidewall of the recess and the first semiconductor chip. The second die structure may include a second semiconductor chip electrically connected to the first semiconductor chip. The first semiconductor chip may include a first circuit substrate, a first front insulating layer formed on a first surface of the first circuit substrate and provided with first bonding pads, and a first backside insulating layer formed on a second surface opposite to the first surface of the first circuit substrate and provided with second bonding pads.
[0016] According to example embodiments, an outer side surface of the second semiconductor chip and an outer side surface of the support substrate may be positioned on the same plane. The first backside insulating layer of the first semiconductor chip may laterally extend along the second surface of the first circuit substrate to cover the gap filling layer. The first backside insulating layer of the first semiconductor chip may be directly bonded to the second front insulating layer of the second semiconductor chip. The second surface of the first circuit substrate of the first semiconductor chip and the second surface of the support substrate that includes silicon may be located on the same plane. The second surface of the first circuit substrate of the first semiconductor chip and the upper surface of the gap filling layer may be located on the same plane.
[0017] According to example embodiments, since no step difference occurs between a backside surface of the first circuit substrate of the first semiconductor chip and the upper surface of the gap filling layer, it may be possible to prevent voids from occurring during a wafer-to-wafer bonding process of the support substrate, in which the first semiconductor chip is formed, and the second semiconductor chip.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0018] Example embodiments will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings. FIGS. **1** to **29** represent non-limiting, example embodiments as described herein.
[0019] FIG. **1** is a cross-sectional view illustrating a semiconductor package in accordance with example embodiments.
[0020] FIG. **2** is an enlarged cross-sectional view illustrating portion ‘A’ in FIG. **1**.
[0021] FIG. **3** is a plan view illustrating a lower die structure in FIG. **1**.
[0022] FIGS. **4** to **19** are views illustrating a method of manufacturing a semiconductor package in accordance with example embodiments.
[0023] FIG. **20** is a cross-sectional view illustrating a semiconductor package in accordance with example embodiments.
[0024] FIG. **21** is a cross-sectional view illustrating an intermediate core die stack of the semiconductor package in FIG. **20**.
[0025] FIGS. **22** to **28** are cross-sectional views illustrating a method of manufacturing a semiconductor package in accordance with example embodiments.
[0026] FIG. **29** is a cross-sectional view illustrating a semiconductor package in accordance with example embodiments.
DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS
[0027] Hereinafter, example embodiments will be explained in detail with reference to the accompanying drawings.
[0028] Ordinal numbers such as “first,” “second,” “third,” etc. may be used simply as labels of certain elements, steps, etc., to distinguish such elements, steps, etc. from one another. Terms that are not described using “first,” “second,” etc., in the specification, may still be referred to as “first” or “second” in a claim. In addition, a term that is referenced with a particular ordinal number (e.g., “first” in a particular claim) may be described elsewhere with a different ordinal number (e.g.,

“second” in the specification or another claim).
[0029] Items described in the singular herein may be provided in plural, as can be seen, for example, in the drawings. Thus, the description of a single item that is provided in plural should be understood to be applicable to the remaining plurality of items unless context indicates otherwise.
[0030] FIG. **1** is a cross-sectional view illustrating a semiconductor package in accordance with example embodiments. FIG. **2** is an enlarged cross-sectional view illustrating portion ‘A’ in FIG. **1**. FIG. **3** is a plan view illustrating a lower die structure in FIG. **1**. FIG. **1** is a cross-sectional view along the line B-B′ of FIG. **3**.
[0031] Referring to FIGS. **1** to **3**, a sub-package **200** may include a first die structure CD**1** having a first semiconductor chip **20** and a second die structure CD**2** stacked on the first die structure CD**1** having a second semiconductor chip **40** that is bonded to the first semiconductor chip **20**. Additionally, the sub-package **200** may further include conductive bumps **50** provided on an outer surface of the first semiconductor chip **20**.
[0032] In example embodiments, the sub-package **200** may be a multi-chip package (MCP) including different types of semiconductor chips. For example, the sub-package **200** may be a chiplet package that includes a plurality of stacked chiplet dies. The stacked semiconductor chips may include the first semiconductor chip **20** as a first chiplet die and the second semiconductor chip **40** as a second chiplet die. For example, the first semiconductor chip **20** and the second semiconductor chip **40** may be small structural units of a semiconductor processor or IP block units that constitute a semiconductor processor. The first semiconductor chip **20** and the second semiconductor chip **40** may be stacked on each other to form a semiconductor device (e.g., a semiconductor processor).
[0033] As used herein, a semiconductor device may refer, for example, to a device such as a semiconductor chip (e.g., memory chip and/or logic chip formed on a die), a stack of semiconductor chips, a semiconductor package including one or more semiconductor chips stacked on a package substrate, or a package-on-package device including a plurality of packages. These devices may be formed using ball grid arrays, wire bonding, through substrate vias, or other electrical connection elements, and may include memory devices such as volatile or non-volatile memory devices. Semiconductor packages may include a package substrate, one or more semiconductor chips, and an encapsulant formed on the package substrate and covering the semiconductor chips.
[0034] The sub-package **200** may be provided as a logic chip including a logic circuit. The logic chip may be a controller that controls a memory chip. For example, the logic chip may be an ASIC serving as a host such as a CPU, GPU, or SOC, or serving as a processor chip such as an application processor AP.
[0035] In some embodiments, the sub-package **200** may include a memory chip such as DRAM, SRAM, etc. For example, the first and second semiconductor chips **20** and **40** may be a memory chip and a logic chip, respectively. The memory chip may be a cache memory chip. Alternatively, in other embodiments, both semiconductor chips **20** and **40** may be memory chips.
[0036] In this embodiment, the sub-package as a multi-chip package is illustrated as including two stacked semiconductor chips (also described as chiplet dies or dies) **20** and **40**. However, the invention is not limited thereto, and for example, the sub-package may include 4, 8, 12, or 16 stacked semiconductor chips.
[0037] In example embodiments, the first die structure CD**1** as a lower die structure may include a support substrate **10** having an opening **16** therein. The first die structure CD**1** may further include a first semiconductor chip **20** disposed in the opening **16**. A gap filling layer **30** may fill a gap G between an inner wall (also described as a sidewall) of the opening **16** and a first semiconductor chip **20**. The second die structure CD**2** as an upper die structure may be stacked on the first die structure CD**1**. The second die structure CD**2** may include a second semiconductor chip **40** electrically and/or directly connected to the first semiconductor chip **20**. The first and second die

structures CD**1** and CD**2** may be subject to certain processes to form a final sub-package. Accordingly, the die structures may be described as intermediate packages.

[0038] It will be understood that when an element is referred to as being “connected” or “coupled” to or “bonded” to or “on” another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to or “directly bonded” to another element, or as “contacting” or “in contact with” another element (or using any form of the word “contact”), there are no intervening elements present at the point of contact.

[0039] As used herein, components described as being “electrically connected” are configured such that an electrical signal can be transferred from one component to the other (although such electrical signal may be attenuated in strength as it is transferred and may be selectively transferred). Moreover, components that are “directly electrically connected” form a common electrical node through electrical connections by one or more conductors, such as, for example, wires, pads, internal electrical lines, through vias, etc. As such, directly electrically connected components do not include components electrically connected through active elements, such as transistors or diodes.

[0040] As illustrated in FIG. **1**, the support substrate **10** may have a first surface (e.g., a lower surface) **12** and a second surface (e.g., an upper surface) **14** opposite to the first surface **12**. The support substrate **10** may have the opening **16** in a center region thereof in a plan view (as viewed from the vertical direction Z). The opening **16** may extend from the first surface **12** to the second surface **14** of the support substrate **10**. For example, the support substrate **10** may be or include at least one of a silicon substrate, a silicon substrate, a non-metallic plate, a metallic plate, etc. A thickness of the support substrate **10** may be within a range of 10 μm to 50 μm.

[0041] The first semiconductor chip **20** may include a first substrate **21** and a first front insulating layer **22**, a plurality of first bonding pads **23**, a plurality of through electrodes **24**, a first backside insulating layer **26** and a plurality of second bonding pads **27**.

[0042] The first substrate **21** may have a first surface **212** and a second surface **214** opposite the first surface **212**. Circuit patterns may be formed on or provided with the first surface **212** of the first substrate **21**. The first surface **212** may be an active surface, and the second surface may be an inactive surface. For example, the first substrate **21** may be a single crystal silicon substrate. The circuit patterns may include transistors, capacitors, diodes, etc. The circuit patterns may constitute circuit elements (e.g., a logic gate, such as a NAND, OR, XOR, NOT (an inverter), NAND, NOR, or an XNOR gate). For example, the first semiconductor chip **20** may be a semiconductor device in which a plurality of circuit elements are formed.

[0043] As illustrated in FIG. **2**, the first front insulating layer **22** may be provided on the first surface **212** of the first substrate **21**. For example, the first surface **212** may be described as an active surface of the first substrate **21**. The first front insulating layer **22** may include a plurality of insulating layers **222** and **224**. A plurality of first metal wiring layers **223** may be provided in the insulating layers. In addition, the first bonding pads **23** may be provided in an outermost insulating layer (e.g., a first passivation layer) **224** of the first front insulating layer **22**.

[0044] For example, the first front insulating layer **22** may surround the plurality of first metal wiring layers **223**. For example, the plurality of first metal wiring layers **223** may be a metal wiring structure having the plurality of wirings vertically stacked in the insulating layer **222**. The first bonding pad **23** may be formed on an uppermost wiring among the plurality of wirings **223**. For example, the wirings may include (or be formed of) aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), platinum (Pt), or an alloy thereof.

[0045] Throughout the specification, when a component is described as “including” (or any form of the word “include”) a particular element or group of elements, it is to be understood that the component is formed of only the element or the group of elements, or the element or group of elements may be combined with additional elements to form the component, unless the context indicates otherwise. The term “consisting of,” on the other hand, indicates that a component is formed only of the element(s) listed.

[0046] The first passivation layer **224** may be formed on the first metal wiring layer **223** and may expose at least a portion of the first bonding pad **23**. The first passivation layer **224** may be a composite layer including a plurality of stacked insulating layers. For example, the first passivation layer **224** may include an oxide layer, silicon nitride or silicon carbonitride. The first passivation layer **224** may have a single-layer or multi-layer structure.

[0047] The first bonding pad **23** may be provided in the first passivation layer **224**. The first bonding pad **23** may be exposed through an opening of the first passivation layer **224** or beyond an outer surface of the first passivation layer **224**. Although not illustrated in the figures, an additional insulation layer may be provided on the first surface **212** of the first substrate **21** to cover the circuit patterns.

[0048] The through electrode **24** such as a through silicon via (TSV) may vertically penetrate the first substrate **21**, and may extend from the first surface **212** to the second surface **214**. Although not illustrated in the figures, in an alternative embodiment, the through electrode **24** may extend from either the first surface **212** or the second surface **214** to a predetermined depth, and the through electrode **24** may not entirely vertically penetrate the first substrate **21**. The through electrode **24** may contact the lowest wiring of the metal wiring structure. Accordingly, the through electrode **24** may be electrically connected to the first bonding pad **23** through the wirings **223**. In another alternative embodiment, the through electrode **24** may entirely vertically penetrate the insulating layer **222**, thereby the through electrode **24** being in contact with the first bonding pad **23**.

[0049] The first backside insulating layer **26** may be formed on the second surface **214** of the first substrate **21**. The second surface **214** may be described as a backside surface of the first substrate **21**. The second bonding pad **27** may be provided in the first backside insulating layer **26**. For example, the second bonding pad **27** may be disposed on an exposed surface of the through electrode **24**. The first backside insulating layer **26** may be formed of or include at least one of silicon oxide, carbon-doped silicon oxide, silicon carbonitride (SiCN), etc. The first bonding pads **23** and the second bonding pad **27** may be electrically connected to each other by the through electrode **24**.

[0050] As illustrated in FIG. **3**, the gap filling layer **30** may fill a gap G between an outer side surface of the first semiconductor chip **20** and a sidewall of the opening **16** of the support substrate **10**. The gap filling layer **30** may surround the first semiconductor chip **20** in a plan view. For example, the gap filling layer **30** may be formed of or include silicon oxide such as TEOS. The gap filling layer may include an inorganic dielectric layer or an organic dielectric layer. The inorganic dielectric layer may be formed of or include at least one of silicon oxide, silicon oxynitride, phosphosilicate glass (PSG), boro-phosphosilicate glass (BPSG), etc. The organic dielectric layer may include a polymer or the like. The gap filling layer may include or be an insulating material having a coefficient of thermal expansion smaller than the support substrate **10**. The gap filling layer may have a hardness lower than the support substrate **10**.

[0051] The first front insulating layer **22** of the first semiconductor chip **20** may be exposed by the gap filling layer **30**. The gap filling layer **30** may fill a gap between an outer side surface of the first front insulating layer **22** and a sidewall of the opening **16** of the support substrate **10**. The gap filling layer **30** may surround the first front insulating layer **22** in a plan view. An outer surface (i.e., a lower surface) of the first front insulating layer **22** and a lower surface of the gap filling layer **30** may be located on the same plane. The lower surface of the gap filling layer **30** and the first surface **12** of the support substrate **10** may be located on the same plane.

[0052] Terms such as “same,” “equal,” “planar,” or “coplanar,” as used herein when referring to orientation, layout, location, shapes, sizes, compositions, amounts, or other measures do not necessarily mean an exactly identical orientation, layout, location, shape, size, composition, amount, or other measure, but are intended to encompass nearly identical orientation, layout, location, shapes, sizes, compositions, amounts, or other measures within typical acceptable variations that may occur resulting from conventional manufacturing processes or within typical acceptable variations that may occur resulting from a noise component or a measurement error when measured. The term “substantially” may be used herein to emphasize this meaning, unless the context or other statements indicate otherwise.

[0053] The second surface **214** of the first substrate **21** of the first semiconductor chip **20** may be coplanar with the second surface **14** of the support substrate **10**. The second surface **214** of the first substrate **21** and the upper surface of the gap filling layer **30** may be located on the same plane. The upper surface of the gap filling layer **30** and the second surface **14** of the support substrate **10** may be located on the same plane.

[0054] The first backside insulating layer **26** of the first semiconductor chip **20** may extend laterally from the second surface **214** of the first substrate **21** to cover the gap filling layer **30** and the second surface **14** of the support substrate **10**.

[0055] In example embodiments, the second semiconductor chip **40** may be stacked on the first semiconductor chip **20** and the support substrate **10**. The second semiconductor chip **40** may include a second substrate **41**, a second front insulating layer **42**, and a plurality of third bonding pads **43**.

[0056] Circuit patterns may be formed on or provided with a first surface **412** of the second substrate **41**, that is an active surface. For example, the second substrate **41** may be a single crystal silicon substrate. The circuit patterns may include transistors, capacitors, diodes, etc. The circuit patterns may constitute circuit elements. The second semiconductor chip **40** may be a semiconductor device in which a plurality of circuit elements are formed.

[0057] The second front insulating layer **42** may be provided on the first surface **412** of the second substrate **41**, that is the active surface. The second front insulating layer **42** may include a plurality of insulating layers **422** and **424**. A plurality of second metal wiring layers **423** may be provided in a first passivation layer **422**. In addition, the third bonding pads **43** may be provided in an outermost insulating layer (e.g., a third passivation layer) **424** of the second front insulating layer **42**.

[0058] For example, the second front insulating layer **42** may surround the plurality of the second metal wiring layers **423**. For example, the plurality of second metal wiring layers **423** may be a metal wiring structure including a plurality of wirings vertically stacked in the insulating layer **422**. The third bonding pad **43** may be formed on an uppermost wiring among the plurality of wirings **423**. For example, the wirings may include aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), platinum (Pt), or an alloy thereof.

[0059] The third passivation layer **424** may be formed on the plurality of second metal wiring layers **423** and may expose at least a portion of the third bonding pad **43**. The third passivation layer **424** may be a composite layer including a plurality of stacked insulating layers. For example, the third passivation layer **424** may include or be formed of silicon oxide, silicon nitride, or silicon carbonitride. The third passivation layer **424** may have a single-layer or multi-layer structure.

[0060] The third bonding pad **43** may be provided in the third passivation layer **424**. The third bonding pad **43** may be exposed through an opening of the third passivation layer **424** or beyond an outer surface of the third passivation layer **424**.

[0061] As illustrated in FIG. **2**, the first semiconductor chip **20** and the second semiconductor chip **40** may be bonded to each other by hybrid bonding. The front surface (e.g., a first surface) **412** of the second substrate (e.g., second circuit substrate) **41** of the second semiconductor chip **40** may face the backside surface **214** (e.g., a second surface) of the first substrate (also described first circuit substrate) **21** of the first semiconductor chip **20**. The second front insulating layer **42** of the second semiconductor chip **40** and the first backside insulating layer **26** of the first semiconductor chip **20** may be directly bonded to each other. In addition, the second bonding pad **27** of the first semiconductor chip **20** and the third bonding pad **43** of the second semiconductor chip **40** may be bonded to each other by Cu—Cu bonding. For example, the second bonding pad **27** of the first semiconductor chip **20** and the third bonding pad **43** of the second semiconductor chip **40** may be directly bonded to each other.

[0062] The first backside insulating layer **26** of the first semiconductor chip **20** and the second front insulating layer **42** of the second semiconductor chip **40** may be directly bonded to each other. The first backside insulating layer **26** and the second front insulating layer **42** may be formed of a material providing excellent bonding strength, thereby contacting each other to form a bonding structure. The first backside insulating layer **26** and the second front insulating layer **42** may be bonded to each other by a high-temperature annealing process while in contact with each other, thereby accomplishing covalent bonding having a relatively strong bonding strength.

[0063] In example embodiments, an outer side surface of the second semiconductor chip **40** and an outer side surface of the support substrate **10** may be located on the same plane. In a plan view (as viewed from the vertical direction Z), the first semiconductor chip **20** may have a first size, and the second semiconductor chip **40** may have a second size that is greater than the first size. The first semiconductor chip **20** may have a first width (e.g., a first length) L**1**, and the second semiconductor chip **40** may have a second width (e.g., a second length) greater than the first width. The first semiconductor chip **20** may be disposed on a center region of the second semiconductor chip **40** in a plan view (as viewed from the vertical direction Z).

[0064] In example embodiments, the support substrate **10** may have a first coefficient of thermal expansion, and the gap filling layer **30** may have a second coefficient of thermal expansion that is greater than the first coefficient of thermal expansion. Since the support substrate **10** containing silicon has a denser composition than the gap filling layer **30**, the support substrate **10** may have excellent mechanical strength and thermal resistance characteristics to temperature changes (or temperature hysteresis) compared to the gap filling layer **30**. When viewed in a plan view, an area ratio occupied by the support substrate **10** to the total area of the first die structure CD may be within a range of 5% to 30%.

[0065] In example embodiments, the conductive bumps **50** may be disposed on the first bonding pads **23** on the first front insulating layer **22** of the first semiconductor chip **20**, respectively. For example, each of the conductive bumps **50** may include a pillar bump **52** on the first bonding pad **23** and a solder bump **54** on the pillar bump **52**. For example, the pillar bump may include copper (Cu), aluminum (Al), tin (Sn), nickel (Ni), gold (Au), platinum (Pt), or an alloy thereof. The solder bump may include solder. The sub-package **200** may be mounted on a substrate such as a package substrate, an interposer, or a redistribution wiring layer via the conductive bumps **50** to form a semiconductor package.

[0066] As mentioned above, the sub-package **200** may include the first die structure CD**1** and the second die structure CD**2** stacked on the first die structure CD**1**. The first die structure CD**1** may include the support substrate **10** having the opening **16** therein, the first semiconductor chip **20** disposed in the opening **16**, and the gap filling layer **30** that fills the gap G between the sidewall of the opening **16** and the first semiconductor chip **20**. The second die structure CD**2** may include the second semiconductor chip **40** that is electrically connected to the first semiconductor chip **20**.

[0067] The outer side surface of the second semiconductor chip **40** and the outer side surface of the support substrate **10** may be located on the same plane. The first backside insulating layer **26** of the first semiconductor chip **20** may extend laterally from the second surface **214** of the first substrate **21** to cover the gap filling layer **30** and the second surface of the support substrate **10**. The first backside insulating layer **26** of the first semiconductor chip **20** and the second front insulating layer **42** of the second semiconductor chip **40** may be directly bonded to each other.

[0068] The second surface **214** of the first substrate **21** of the first semiconductor chip **20** and the second surface **14** of the support substrate **10** that includes silicon may be located on the same plane. The second surface **214** of the first substrate **21** of the first semiconductor chip **20** and the

upper surface of the gap filling layer **30** may be located on the same plane. The upper surface of the gap filling layer **30** and the second surface **14** of the support substrate **10** may be located on the same plane. Accordingly, since no step difference occurs between the backside surface of the first substrate **21** of the first semiconductor chip **20** and the upper surface of the gap filling layer **30**, it may be possible to prevent voids from occurring during a bonding process of the wafers in which the first semiconductor chip **20** and the second semiconductor chip **40** are formed, respectively.
[0069] Hereinafter, a method of manufacturing the semiconductor package of FIG. **1** will be described.
[0070] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” “top,” “bottom,” and the like, may be used herein for ease of description to describe one element's or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.
[0071] FIGS. **4** to **19** are views illustrating a method of manufacturing a sub-package in accordance with example embodiments. FIGS. **4**, **6**, **9** to **12**, **14**, and **16** to **19** are cross-sectional views illustrating a method of manufacturing a sub-package in accordance with example embodiments. FIG. **7** is an enlarged cross-sectional view illustrating portion ‘D’ in FIG. **6**. FIG. **13** is an enlarged cross-sectional view illustrating portion ‘F’ in FIG. **12**. FIG. **15** is an enlarged cross-sectional view illustrating portion ‘G’ in FIG. **14**. FIG. **5** is a plan view of FIG. **4**. FIG. **4** is a cross-sectional view taken along the line C-C′ in FIG. **5**. FIG. **8** is a plan view of FIG. **6**. FIG. **6** is a cross-sectional view taken along the line E-E′ in FIG. **8**.
[0072] Referring to FIGS. **4** and **5**, a carrier substrate C**1** including a recess RC formed therein may be prepared. In example embodiments, the carrier substrate C**1** may be provided as a support frame on which a first semiconductor chip (e.g., lower dies) is placed in the recess RC.
[0073] As described later, a gap filling layer may be formed to fill the recess, and the first semiconductor chip may be bonded to a second semiconductor chip of a target substrate (e.g., the second wafer W**2** in FIG. **14**) by using a substrate-to-substrate bonding. On the target substrate, a plurality of the second semiconductor chips **40** (shown in FIG. **1**) may be formed.
[0074] The carrier substrate C**1** and the target substrate have the same shape as each other in a plan view (as viewed from the vertical direction Z). For example, the carrier substrate C**1** and the target substrate may have a shape of a wafer or a panel. For example, the carrier substrate C**1** may be or include at least one of a silicon substrate, a silicon substrate, a non-metallic plate, a metallic plate, etc.
[0075] The carrier substrate C**1** may be or include a silicon substrate **10** that has a first surface **12** and a second surface **14** opposite to the first surface **12**. The carrier substrate C**1** may include a package region PR on which the first semiconductor chip is mounted and a cutting region CR surrounding the package region PR. As will be described later, the carrier substrate C**1** may be cut along the cutting region CR to form a plurality of individual support substrates. The cutting region CR may surround the first semiconductor chip.
[0076] As illustrated in FIGS. **4** and **5**, alignment key patterns AL may be formed on the second surface **14** of the carrier substrate C**1**. For example, an oxide layer may be formed on the second surface **14** of the silicon substrate **10**, and a patterning process and a plating process may be performed on the oxide layer to form the alignment key patterns AL. The alignment key patterns AL may include a metal material such as aluminum or copper. The alignment key patterns AL may be formed in a peripheral region of the package region PR or in the cutting region CR. As described

later, a process of attaching the first semiconductor chip may be performed using the alignment key patterns AL.

[0077] The recess RC may be formed in the second surface **14** of the carrier substrate C**1**. The recess RC may accommodate the first semiconductor chip in a subsequent process step. The recesses RC may be formed in the package regions PR of the carrier substrate C**1**. For each of the package regions PR of the carrier substrate C**1**, at least one of the alignment key patterns AL may be used during the attaching process. The recess RC may be formed by a plasma etching process. Alternatively, the recess RC may be formed by a laser drilling process, a wet etching process, etc.

[0078] The recess RC may be a region in which the first semiconductor chip is disposed. The predetermined depth DT of the recess RC may be within a range of 10 μm to 50 μm. The recess RC may have an area greater than an area of the first semiconductor chip in a plan view (as viewed from the vertical direction Z). The recess RC may have a predetermined depth DT from the second surface **14** of the silicon substrate **10**. For example, the area of the recess RC may be within a range of 105% to 115% of the area of the first semiconductor chip in a plan view (as viewed from the vertical direction Z). The area and depth of the recess RC may be determined in consideration of the area and thickness of the first semiconductor chip.

[0079] In some embodiments, the alignment key patterns AL may be formed within the recess RC. In this case, an oxide layer as a mold pattern layer may be formed on a bottom surface of the recess RC of the silicon substrate **10**, and a patterning process and a plating process may be formed on the mold pattern layer to form the alignment key patterns AL.

[0080] Referring to FIGS. **6** to **8**, a first semiconductor chip **20** may be disposed in the recess RC of the silicon substrate **10**.

[0081] In example embodiments, a plurality of first semiconductor chips **20** may be provided. For example, the plurality of first semiconductor chips **20** may be individualized from a wafer by a sawing process. The plurality of first semiconductor chips **20** may be placed in the recesses RC of the silicon substrate **10**. A front surface of the first semiconductor chip **20** may be stacked to face the silicon substrate **10**. An upper portion of the first semiconductor chip **20** may protrude from the recess RC beyond the second surface **14** of the silicon substrate **10**. The first semiconductor chip **20** may be a first chiplet die (e.g., a lower chiplet die). The first semiconductor chip **20** may be a small structural unit of a semiconductor processor or IP block units that constitute a semiconductor processor. The first semiconductor chip **20** and the second semiconductor chip **40** may be stacked on each other to form a semiconductor device (e.g., a semiconductor processor).

[0082] As illustrated in FIG. **7**, the first semiconductor chip **20** may include a first substrate **21** and a first front insulating layer **22** provided with first bonding pads **23**. Additionally, the first semiconductor chip **20** may further include a plurality of through electrodes **24** that are provided in the first substrate **21** and are electrically connected to the first bonding pads **23**.

[0083] The first substrate **21** may have a first surface **212** and a second surface **214** opposite the first surface **212**. Circuit patterns may be formed on the first surface **212** of the first substrate **21**. For example, the first substrate **21** may include silicon, germanium, silicon-germanium, or III-V compounds (e.g., GaP, GaAs, GaSb, etc.). In some embodiments, the first substrate **21** may be a silicon-on-insulator (SOI) substrate, or a germanium-on-insulator (GOI) substrate.

[0084] The circuit patterns may include transistors, capacitors, diodes, etc. The circuit patterns may constitute circuit elements. Accordingly, the first semiconductor chip may be a semiconductor device in which a plurality of circuit elements are formed. The circuit patterns may be formed by performing a wafer fabrication process (which may be a Front End of Line (FEOL) process) on the first surface **212** of the first substrate **21**. A surface of the first substrate, on which the FEOL process is performed, may be described as a front surface of the first substrate. A surface opposite to the front surface may be referred to as a backside surface.

[0085] The first front insulating layer **22** as an insulation layer may be formed on the first surface (front surface) **212** of the first substrate. The first front insulating layer **22** may include a plurality of insulating layers **222** and **224**. A plurality of first metal wiring layers **223** may be provided in the insulating layers. In addition, the first bonding pads **23** may be provided in an outermost insulating layer (e.g., a first passivation layer) **224** of the first front insulating layer **22**.

[0086] For example, the first front insulating layer **22** may surround the plurality of first metal wiring layer **223** and a first passivation layer **224**. For example, the plurality of first metal wiring layer **223** may be a metal wiring structure having the plurality of wirings vertically stacked in the insulating layer **222**. The first bonding pad **23** may be formed on an uppermost wiring among the plurality of wirings **223**. For example, the wirings may include (or be formed of) aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), platinum (Pt), or an alloy thereof.

[0087] The first passivation layer **224** may be formed on the first metal wiring layer **223** and may expose at least a portion of the first bonding pad **23**. The first passivation layer **224** may be a composite layer including a plurality of stacked insulating layers. For example, the first passivation layer **224** may include an oxide layer, silicon nitride or silicon carbonitride. The first passivation layer **224** may have a single-layer or multi-layer structure.

[0088] The first bonding pad **23** may be provided in the first passivation layer **224**. The first bonding pad **23** may be exposed through an opening of the first passivation layer **224** or beyond an outer surface of the first passivation layer **224**. Although not illustrated in the figures, an additional insulation interlayer may be provided on the first surface **212** of the first substrate **21** to cover the circuit patterns.

[0089] The insulation layer **22** may be formed of, for example, silicon oxide or a low dielectric material. The first metal wiring layer **223** may be electrically connected to the circuit patterns. Accordingly, the circuit pattern may be electrically connected to the first bonding pad **23** by the first metal wiring layer **223**.

[0090] The through electrode **24** such as through silicon via (TSV) may extend from the first surface **212** of the first substrate **21** to a predetermined depth. The through electrode **24** may contact the first metal wiring layer **223**. Accordingly, the through electrode **24** may be electrically connected to the first bonding pad **23** through the first metal wiring layer **223**. In some embodiments, the substrate **21** may further include an insulating layer and the through electrode **24** may vertically penetrate the insulation interlayer.

[0091] A liner layer (not illustrated in the drawings) may be provided on an outer surface of the through electrode **24**. The liner layer may include or be formed of silicon oxide or carbon-doped silicon oxide. The liner layer may electrically insulate the through electrode **24** from the first substrate **21** and the first metal wiring layer **223**.

[0092] The through electrode **24** and the first bonding pad **23** may include the same metal. For example, the metal may include copper (Cu). However, the invention is not limited thereto, and the metal may further include a material (e.g., gold (Au)) that can be combined by interdiffusion of metals (e.g., interdiffusion between Au and Cu) by a high-temperature annealing process.

[0093] As illustrated in FIG. **8**, an outer side surface of the first semiconductor chip **20** may be spaced apart from a sidewall of the recess RC. A gap G between the outer side surface of the first semiconductor chip **20** and the sidewall of the recess RC may be within a range of 5 μm to 10 μm.

[0094] Referring to FIG. **9**, the second surface **214** of the first substrate **21** may be partially removed to expose an end portion of the through electrode **24**.

[0095] In example embodiments, a grinding process such as a back lap process may be performed to partially remove a portion of the first substrate **21**, which is adjacent to the second surface **214** of the first substrate **21**. In the back lap process, the entire backside surface of the first substrate **21** may be subject to the grinding process (i.e., the back lap process).

[0096] Subsequently, a silicon recess process such as an etching process may be performed to further remove an additional portion of the first substrate **21**, which is adjacent to the second surface **214** of the first substrate **21** and to expose end portions of the through electrodes **24**. Accordingly, a thickness of the first substrate **21** may be reduced to a desired thickness. For example, the thickness of the first substrate **21** may be in a range of from about 20 μm to about 100 μm. As a result of the silicon recess process, the second surface **214** of the first substrate **21** may be coplanar or higher than the second surface **14** of the silicon substrate **10**. The silicon recess process may selectively etch only silicon in the backside surface of the first substrate **21** without removing the silicon in the second surface **14** of the silicon substrate **10**. Accordingly, the end portions of the through electrodes **24** may protrude beyond the second surface **214** of the first substrate **21**. The etching process may be an isotropic dry etching process. The etching process may include a plasma etching process. The plasma etching process may be performed using inductively coupled plasma, capacitively coupled plasma, microwave plasma, etc. For example, the protruding end portions of the through electrodes **24** may have the same heights from the second surface **214** of the first substrate **21** as each other.

[0097] In an alternative embodiment, the back lap process may be performed to expose the end portions of the through electrodes **24**, and optionally the silicon recess process may not be performed. In another alternative embodiment, the back lap process may be performed to expose the end portions of the through electrodes **24** such that the end portions protrudes beyond the second surface **214** of the first substrate **21**, and optionally the silicon recess process may not be performed.

[0098] Referring to FIG. **10**, a gap filling layer **30** may be formed on the second surface **14** of the carrier substrate C**1** to cover the first semiconductor chip **20**.

[0099] In example embodiments, the gap filling layer **30** may be formed to fill the gap between the outer side surface of the first semiconductor chip **20** and the sidewall of the recess RC. The gap filling layer **30** may be formed to cover the second surface **14** of the silicon substrate **10**. The gap filling layer **30** may be formed to fill the gap between the protruding end portions of the through electrodes **24** on the second surface **214** of the first substrate **21**. The gap filling layer **30** may be formed of or include silicon oxide such as TEOS.

[0100] Referring to FIGS. **11** to **13**, a first backside insulating layer **26** provided with second bonding pads **27** in an outer surface thereof may be formed on the second surface **214** of the first substrate **21**.

[0101] As illustrated in FIG. **11**, an upper portion of the gap filling layer **30** may be removed to expose the end portions of the through electrodes **24**. At this time, the second surface **214** of the first substrate **21** may be exposed as well. For example, the second surface **214** of the first substrate **21** may be exposed simultaneously with the end portions of the through electrodes **24**.

[0102] In example embodiments, a chemical mechanical polishing (CMP) process may be performed to remove the upper portion of the gap filling layer **30** to expose the end portions of the through electrodes **24**. For example, during the CMP process, the second surface **14** of the silicon substrate **10** (e.g., a passivation layer on the second surface **14**) may be utilized to detect a polishing end point. Through the CMP process, the end portions of the through electrodes **24** as well as the portion of the gap filling layer covering the end portions may be removed to expose the second surface **214** of the first substrate **21** (e.g., the passivation layer on the second surface **14**). The second surface **14** of the silicon substrate **10** and the second surface **214** of the first substrate **21** may be positioned on the same plane. The second surface **14** of the silicon substrate **10** and an upper surface of the gap filling layer **30** may be positioned on the same plane. The upper surface of the gap filling layer **30** and the second surface **214** of the first substrate **21** may be positioned on the same plane. In some embodiments of the invention, during the CMP process, an upper portion of the second surface **214** of the first substrate **21** may be removed as well.

[0103] Accordingly, since no step difference is generated between the backside surface of the first substrate **21** of the first semiconductor chip **20** and the upper surface of the gap filling layer **30**, voids may be prevented from occurring during a subsequent wafer-to-wafer bonding process of the first semiconductor chip **20** and the second semiconductor chip. For example, since no bevel step difference occurs between the second surface **14** of the silicon substrate **10** and the backside surface of the first substrate **21** of the first semiconductor chip **20** at an edge region of the carrier substrate C**1**, voids may be prevented from occurring in the bevel region during the subsequent wafer-to-wafer bonding process of the first semiconductor chip **20** and the second semiconductor chip.

[0104] As illustrated in FIGS. **12** and **13**, the first backside insulating layer **26** as a second passivation layer may be formed on the second surface **214** of the first substrate **21**, and the second bonding pad **27** that is electrically connected to the through electrode **24** may be formed in the first backside insulating layer **26**.

[0105] For example, the first backside insulating layer **26** may be formed of or include at least one of silicon oxide, carbon-doped silicon oxide, silicon carbonitride (SiCN), etc. After the first backside insulating layer **26** is formed on the second surface **214** of the first substrate **21**, a through hole may be formed in the first backside insulating layer **26** to expose the through electrode **24** and a plating process may be performed to form the second bonding pad **27**. The second bonding pad **27** may be disposed on the exposed upper surface of the through electrode **24**. Accordingly, the first bonding pads **23** and the second bonding pads **27** may be electrically connected to each other by the through electrode **24**.

[0106] Referring to FIGS. **14** and **15**, a second wafer W**2**, in which a second semiconductor chip **40** is formed, may be provided. Subsequently, the carrier substrate C**1** of FIG. **12**, in which the first semiconductor chip **20**, is formed may be attached on the second wafer W**2**, for example, by a wafer-to-wafer hybrid bonding process.

[0107] In example embodiments, the structure of FIG. **12** may be turned over, and the carrier substrate C**1** may be bonded onto the second wafer W**2**. Accordingly, a backside surface of the first semiconductor chip **20** may face a front surface of the second semiconductor chip **40** of the second wafer W**2**.

[0108] When the carrier substrate C**1** and the second wafer W**2** are bonded to each other in a wafer-to-wafer bonding manner, a thermal compression process may be performed. For example, the stacked wafers of the carrier substrate C**1** and the second wafer W**2** may be subject to the thermal compression process such that the first semiconductor chip **20** and the second semiconductor chip **40** are bonded to each other in a manner of hybrid bonding. For example, the second bonding pad **27** may be directly bonded to the third bonding pads **43**. Additionally, the front surface of the second semiconductor chip **40** (i.e., a second front insulating layer **42**) may be directly bonded to the backside surface of the first semiconductor chip **20** (i.e., the backside insulating layer **26**).

[0109] When the carrier substrate C**1** and the second wafer W**2** are bonded to each other by wafer-to-wafer bonding, there is no step difference between the backside surface of the first substrate **21** of the first semiconductor chip **20** and the upper surface of the gap filling layer **30**. Accordingly, it may be possible to prevent voids from occurring between the first backside insulating layer **26** of the first semiconductor chip **20** and the second front insulating layer **42** of the second semiconductor chip **40**. For example, when the carrier substrate C**1** and the second wafer W**2** are bonded to each other by wafer-to-wafer bonding, there is no bevel step difference between the second surface **14** of the silicon substrate **10** and the backside surface of the first substrate **21** of the first semiconductor chip **20** at the edge region of the carrier substrate C**1**. Accordingly, it may be possible to prevent a void from occurring between the edge region of the carrier substrate C**1** and an edge region of the second wafer W**2**.

[0110] As illustrated in FIG. **15**, the second semiconductor chip **40** may include the second substrate **41** and the second front insulating layer **42** provided with third bonding pads **43** in an outer surface thereof. The second substrate **41** may have a first surface **412** and a second surface **414** opposite the first surface **412**. Circuit patterns may be formed on the first surface **412** of the second substrate **41**.

[0111] The second front insulating layer **42** as an insulation layer may be formed on the first surface (front surface) **412** of the second substrate **41**. The second front insulating layer **42** may

include a plurality of insulating layers **422** and **424**. A plurality of second metal wiring layers **423** may be provided in a first passivation layer **422**. In addition, the third bonding pads **43** may be provided in an outermost insulating layer of the second front insulating layer **42**.
[0112] For example, the second front insulating layer **42** may surround the plurality of the second metal wiring layers **423**. For example, the plurality of second metal wiring layers **423** may be a metal wiring structure including a plurality of wirings vertically stacked in the insulating layer **422**. The third bonding pad **43** may be formed on an uppermost wiring among the plurality of wirings **423**. For example, the wirings may include aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), platinum (Pt), or an alloy thereof.
[0113] The third passivation layer **424** may be formed on the plurality of second metal wiring layers **423** and **423** and may expose at least a portion of the third bonding pad **43**. The third passivation layer **424** may be a composite layer including a plurality of stacked insulating layers. For example, the third passivation layer **424** may include or be formed of silicon oxide, silicon nitride, or silicon carbonitride. The third passivation layer **424** may have a single-layer or multi-layer structure.
[0114] The third bonding pad **43** may be provided in the third passivation layer (also described as an insulating layer) **424**. The third bonding pad **43** may be exposed through an opening of the third passivation layer **424** or beyond an outer surface of the third passivation layer **424**. Although not illustrated in the figures, an additional insulation layer may be provided on the first surface **412** of the second substrate **41** to cover the circuit patterns.
[0115] The insulation layer **42** may be formed to include, for example, silicon oxide or a low dielectric material. The second metal wiring layer **423** may be electrically connected to the circuit patterns. Accordingly, the circuit pattern may be electrically connected to the third bonding pad **43** through the second metal wiring layer **423**.
[0116] The second semiconductor chip **40** may be a second chiplet die (e.g., an upper chiplet die). The second semiconductor chip **40** may be a small structural unit or IP block unit of the processor chip. The first semiconductor chip **20** and the second semiconductor chip **40** may be stacked on each other to form a semiconductor device (e.g., a semiconductor processor).
[0117] Referring to FIG. **16**, a portion of the carrier substrate C**1** may be partially removed until the front surface of the first semiconductor chip **20** is exposed. The removed portion may be adjacent to the first surface **12** of the carrier substrate C**1**.
[0118] In example embodiments, the first surface **12** of the carrier substrate C**1** may be partially removed by a grinding process. The first surface **12** of the carrier substrate C**1** may be removed to expose the bottom surface of the recess RC. As a result, the recess RC may become an opening **16** that penetrates entirely through the silicon substrate **10**. The first substrate **21** and the first front insulating layer **22** may be disposed within the opening **16** of the silicon substrate **10**. The first front insulating layer **22** of the first semiconductor chip **20** may be exposed. Accordingly, the first bonding pads **23** may be exposed from the first surface **12** of the silicon substrate **10**.
[0119] Accordingly, the first semiconductor chip **20** may be disposed in the opening **16** of the silicon substrate **10**, and the gap filling layer **30** may be formed to fill the gap between the outer side surface of the first semiconductor chip **20** and the sidewall of the opening **16**.
[0120] Referring to FIGS. **17** and **18**, conductive bumps **50** may be formed on the first bonding pads **23** of the first semiconductor chip **20**, respectively.
[0121] Though not shown in the drawings, a seed layer and a photoresist layer may be formed on the first front insulating layer **22**. Subsequently an exposure process may be performed to form a photoresist pattern PL that exposes bump regions. The photoresist pattern PL may be provided with through holes OP, as illustrated in FIG. **17**.
[0122] Though not shown in the drawings, after filling the openings OR of the photoresist pattern PL with a conductive material, the photoresist pattern may be removed. The conductive material may be a composite layer including a pillar bump layer and a solder bump layer.

[0123] For example, a first plating process may be performed to form a pillar bump layer on the first bonding pad **23** of the first semiconductor chip **20**, and a second plating process may be performed to form a solder bump layer on the pillar bump layer. Subsequently, a reflow process may be performed to form the conductive bumps **50** including a pillar bump **52** and a solder bump **54**, as shown in FIG. **18**. Alternatively, the conductive bumps may be formed by one of a screen printing process, a deposition process, etc.

[0124] Referring to FIG. **19**, the bonded structure of the carrier substrate C**1** and the second wafer W**2** may be cut along the cutting region CR to form a plurality of sub-packages **200** as chiplet packages. The sub-package **200** may include the first semiconductor chip (e.g., a first chiplet die) **20** and the second semiconductor chip (e.g., a second chiplet die) **40**, which are bonded to each other.

[0125] Because the cutting process is performed to separate the carrier substrate C**1** and the second wafer W**2** at the same time, an outer side surface of the second semiconductor chip **40** and an outer side surface of the silicon substrate **10** may be positioned on the same plane. For example, an outer side surface of the second semiconductor chip **40** and an outer side surface of the silicon substrate **10** may be coplanar.

[0126] FIG. **20** is a cross-sectional view illustrating a sub-package **500** in accordance with example embodiments. FIG. **21** is a cross-sectional view illustrating an intermediate core die stack of the sub-package in FIG. **20**.

[0127] The sub-package **500** may include an intermediate core die stack, a buffer die BD and a top core die TD. Except for the buffer die BD, the sub-package **500** may be substantially the same as or similar to the sub-package **200** described with reference to FIG. **1**. The intermediate core die stack in FIG. **20** may have substantially the same structure of a stack of three pieces of the second die structure CD**2** in FIG. **1**. The top core die TD in FIG. **20** may have substantially the same structure of the first die structure CD**1** in FIG. **1**. Thus, same or similar reference numerals will be used to refer to the same or like elements and any further repetitive explanation concerning the same or like elements will be omitted.

[0128] Referring to FIGS. **20** and **21**, a sub-package **500** may include semiconductor chips (also described as dies) stacked therein. The sub-package **500** may include a buffer die **60**, and an intermediate core die stack, and a top core die TD stacked on the intermediate core die stack. The intermediate core die stack may include a plurality of die structures DS**1**, DS**2** and DS**3** sequentially stacked. Each of the die structures DS**1**, DS**2** and DS**3** may include a support substrate **10***a*, **10***b* and **10***c* having an opening **16***a*, **16***b* and **16***c* therein, a core die **20***a*, **20***b* and **20***c* disposed in the opening **16***a*, **16***b* and **16***c*, and a gap filling layer **30***a*, **30***b* and **30***c* that fills a gap between a sidewall of the opening and the core die **20***a*, **20***b* and **20***c*. For example, the core die **20***a*, **20***b* and **20***c* may be a semiconductor chip. The plurality of die structures DS**1**, DS**2** and DS**3** sequentially stacked. Each of the plurality of die structures DS**1**, DS**2** and DS**3** may include a corresponding one of the plurality of core dies **20***a*, **20***b* and **20***c* sequentially stacked on one another. The gap filling layers **30***a*, **30***b* and **30***c* may cover side surfaces of the plurality of core dies **20***a*, **20***b* and **20***c*, **20***d* respectively. Additionally, the sub-package **500** may further include conductive bumps **70** provided on an outer surface of the buffer die **60**.

[0129] The intermediate core die stack may include a plurality of semiconductor chips (also described as dies) **20***a*, **20***b* and **20***c* stacked vertically. In this embodiment, the semiconductor chips **20***a*, **20***b* and **20***c* may be substantially the same as or similar to each other. Accordingly, same or like reference numerals will be used to refer to the same or like elements and repeated descriptions of the same elements may be omitted.

[0130] In this embodiment, the sub-package **500** as a multi-chip package is illustrated as including four semiconductor chips **20***a*, **20***b*, **20***c* and **40** stacked on the buffer die **60**, however the invention is not limited thereto. For example, the sub-package may include 8, 12 or 16 semiconductor chips stacked on the buffer die **60**.

[0131] Each of the semiconductor chips **20***a*, **20***b* and **20***c* and **60** may include an integrated circuit. The integrated circuit may be formed by performing wafer-level semiconductor manufacturing processes. Each semiconductor chip **20***a*, **20***b* and **20***c* and **60** may be, for example, a memory chip or a logic chip.
[0132] In an embodiment, the sub-package **500** may be a memory device. For example, the memory device may be a high bandwidth memory (HBM) device. Each of the semiconductor chips **40**, **20***a*, **20***b* and **20***c* may be a semiconductor memory chip, and the buffer die **60** may be a semiconductor buffer chip having a function to control access to the semiconductor memory chips, buffer or convert signals from/into the semiconductor memory chips, or perform any other suitable functions. For example, each of the semiconductor memory chips **40**, **20***a*, **20***b* and **20***c* may have the same storage capacity as each other.
[0133] As illustrated in FIG. **21**, the intermediate core die stack may include first, second and third die structures DS**1**, DS**2** and DS**3** sequentially stacked. The first, second and third die structures DS**1**, DS**2** and DS**3** may have the same structure as each other. The first die structure DS**1** may include a support substrate **10***a* having an opening **16***a*, a first core die **20***a* disposed in the opening **16***a*, and a gap filling layer **30***a* filling a gap between a sidewall of the opening **16***a* and the first core die **20***a*. The second die structure DS**2** may include a support substrate **10***b* having an opening **16***b*, a second core die **20***b* disposed in the opening **16***b*, and a gap filling layer **30***b* filling a gap between a sidewall of the opening **16***b* and the second core die **20***b*. The third die structure DS**3** may include a support substrate **10***c* having an opening **16***c*, a third core die **20***c* disposed in the opening **16***c*, and a gap filling layer **30***c* filling a gap between a sidewall of the opening **16***c* and the third core die **20***d*. The first, second, and third die structures DS**1**, DS**2**, and DS**3** may be substantially the same as or similar to the first die structure CD**1** of FIG. **1**. Accordingly, same or like reference numerals will be used to refer to the same or like elements and repeated descriptions of the same elements may be omitted.
[0134] In example embodiments, the buffer die **60** may include a substrate **61**, a front insulating layer **62**, a plurality of first bonding pads **63**, a plurality of through electrodes **64**, and a backside insulating layer **66**, and a plurality of second bonding pads **67**. Additionally, the buffer die **60** may further include the conductive bumps **70** as conductive connection members provided respectively on the first bonding pads **63**. The buffer die **60** may be mounted on a package substrate or an interposer via the conductive bumps **70**. For example, the conductive bump **70** may include a pillar bump **72** on the first bonding pad **63** and a solder bump **74** on the pillar bump **72**.
[0135] In example embodiments, the first, second and third die structures DS**1**, DS**2** and DS**3** may be sequentially stacked on the buffer die **60**. The top die TD may be stacked on the third die structure DS**3**.
[0136] The first die structure DS**1** may be bonded to a wafer including the buffer die **60** by hybrid bonding during a manufacturing process. The second bonding pad **67** of the buffer die **60** and a first bonding pad **23***a* of the first core die **20***a* may be bonded to each other by Cu—Cu bonding. A front surface of the first core die **23***a* (i.e., a first front insulating layer **22***a* on a first surface **212***a* of a first substrate (also described first circuit substrate) **21***a*) may be directly bonded to the backside insulating layer **66** of the substrate (also described circuit substrate) **61** of the buffer die **60**.
[0137] The second die structure DS**2** may be bonded to the first die structure DS**1** by hybrid bonding. A second bonding pad **27***a* of the first core die **20***a* and a first bonding pad **23***b* of the second core die **20***b* may be bonded to each other by Cu—Cu bonding.
[0138] A first front insulating layer **22***b* of the second core die **20***b* and a first backside insulating layer **26***a* of the first core die **20***a* may be directly bonded to each other. The first backside insulating layer **26***a* and the first front insulating layer **22***b* may make contact with each other to provide a bonding structure. The first backside insulating layer **26***a* and the first front insulating layer **22***b* may include an insulating material providing excellent bonding strength. The first backside insulating layer **26***a* and the first front insulating layer **22***b* may be bonded to each other

by a high-temperature annealing process while in contact with each other, thereby accomplishing covalent bonding having a relatively strong bonding strength.

[0139] Similarly, the third die structure DS**3** may be bonded to the second die structure DS**2** by hybrid bonding. A second bonding pad **27***b* of the second core die **20***b* and a first bonding pad **23***c* of the third core die **20***c* may be bonded to each other by Cu—Cu bonding. A first front insulating layer **22***c* of the third core die **20***c* and a first backside insulating layer **26***b* of the second core die **20***b* may be directly bonded to each other.

[0140] The top die TD may be bonded to the third die structure DS**3** by hybrid bonding. A second front insulating layer **42** of the top die TD and the first backside insulating layer **26***c* of the third core die **20***c* may be directly bonded to each other. A second bonding pad **27***c* of the third core die **20***c* and a third bonding pad **43** of the top die TD may be bonded to each other by Cu—Cu bonding (e.g., pad to pad direct bonding).

[0141] Hereinafter, a method of manufacturing the sub-package of FIG. **20** will be described.

[0142] FIGS. **22** to **28** are cross-sectional views illustrating a method of manufacturing a sub-package in accordance with example embodiments. FIG. **23** is an enlarged cross-sectional view illustrating portion 'H' in FIG. **22**.

[0143] Referring to FIGS. **22** and **23**, processes the same as or similar to the processes described with reference to FIGS. **4** to **16** may be performed thereby attaching a first carrier substrate C**1** including third core dies **20***c* formed therein to a second wafer W**2** including top core dies **40** formed therein. Subsequently, processes the same as or similar to the processes described with reference to FIGS. **4** to **13** may be performed, thereby forming a second carrier substrate C**2** including second core dies **20***b* formed therein, and the second carrier substrate C**2** may be attached to a first carrier substrate C**1** by wafer-to-wafer hybrid bonding process. In example embodiments, in view of the manufacturing process, the first semiconductor chip **20** and the second semiconductor chip **40** of FIG. **16** may correspond to the third core die **20***c* and the top core die **40** of FIG. **22**, respectively. By the bonding, a backside surface of the second core die **20***b* of the second carrier substrate C**2** may face a front surface of the third core die **20***c* of the first carrier substrate C**1**.

[0144] When the second carrier substrate C**2** and the first carrier substrate C**1** are bonded to each other by wafer-to-wafer bonding (e.g., by a thermal compression process), the second core die **20***b* of the second carrier substrate C**2** and the third core die **20***c* of the first carrier substrate C**1** may be bonded to each other in a manner of hybrid bonding. For example, as illustrated in FIG. **23**, the bonding pad **27***b* of the second core die **20***b* and the bonding pad **23***c* of the third core die **20***c* may be directly bonded to each other by Cu—Cu bonding. Additionally, the front surface of the third core die **20***c* (i.e., a first front insulating layer **22***a* on a first surface **212***a* of a first substrate (also described circuit substrate) **21***c*) may be directly bonded to the backside surface of the second core die **20***b* (i.e., a backside insulating layer **26***b* on a second surface **214***b* of a first substrate (also described first circuit substrate) **21***b*).

[0145] When the second carrier substrate C**2** and the first carrier substrate C**1** are bonded to each other by wafer-to-wafer bonding, no step difference may be generated between the backside surface of the first substrate **21***b* of the second core die **20***b* and an upper surface of a gap filling layer **30***b*. Accordingly, it may be possible to prevent voids from occurring between the first backside insulating layer **26***b* of the second core die **20***b* and the first front insulating layer **22***c* of the third core die **20***c*. For example, when the second carrier substrate C**2** and the first carrier substrate C**1** are bonded to each other by wafer-to-wafer bonding, no bevel step difference is generated between a second surface **14***b* of a silicon substrate **10***b* and a backside surface of the first substrate **21***b* of the second core die **20***b* at an edge region of the second carrier substrate C**2**. Accordingly, it may be possible to prevent a void from occurring between the edge region of the second carrier substrate C**2** and an edge region of the first carrier substrate C**1**.

[0146] Referring to FIG. **24**, a first surface **12***b* of the second carrier substrate C**1** may be partially removed until a front surface of the second core die **20***b* is exposed.

[0147] In example embodiments, the first surface **12***b* of the second carrier substrate C**2** may be partially removed by a grinding process. The first surface **12***b* of the second carrier substrate C**2** may be removed to expose a bottom surface of a recess RC**2**. Accordingly, the recess RC**2** may become an opening **16***b* that penetrates the silicon substrate **10***b*. The first substrate **21***b* and the first front insulating layer **22***b* may be disposed within the opening **16***b* of the silicon substrate **10***b*. The first front insulating layer **22***b* of the second core die **20***b* may be exposed by the opening **16***b*. Accordingly, the first bonding pads **23***b* may be exposed from the first surface **12***b* of the silicon substrate **10***b*.

[0148] The second core die **20***b* may be disposed in the opening **16***b* of the silicon substrate **10***b*, and the gap filling layer **30***b* may be formed to fill a gap between an outer side surface of the second core die **20***b* and a sidewall the opening **16***b*.

[0149] Referring to FIG. **25**, processes the same as or similar to the processes described with reference to FIGS. **4** to **13** may be performed to form a third carrier substrate C**3** including first core dies **20***a* formed therein, and the third carrier substrate C**3** may be attached to the second carrier substrate C**2** by wafer-to-wafer hybrid bonding process. In example embodiments, by the bonding, a backside surface of the first core die **20***a* of the third carrier substrate C**3** may face a front surface of the second core die **20***b* of the second carrier substrate C**2**.

[0150] When the third carrier substrate C**3** and the second carrier substrate C**2** are bonded to each other by wafer-to-wafer bonding (e.g., by a thermal compression process), the first core die **20***a* of the third carrier substrate C**3** and the second core die **20***b* of the second carrier substrate C**2** may be bonded to each other in a manner of hybrid bonding. For example, the bonding pad **27***a* of the first core die **20***a* and the bonding pad **23***b* of the second core die **20***b* may be directly bonded to each other by Cu—Cu bonding. Additionally, the front surface of the second core die **20***b* (i.e., a first front insulating layer **22***b* on a first surface **212***b* of a first substrate **21***b*) may be directly bonded to the backside surface of the first core die **20***a* (i.e., a backside insulating layer **26***a* on a second surface **214***a* of a first substrate **21***a*).

[0151] When the third carrier substrate C**3** and the second carrier substrate C**2** are bonded to each other by wafer-to-wafer bonding, no step difference is generated between the backside surface of the first substrate **21***a* of the first core die **20***a* and an upper surface of a gap filling layer **30***a*. Accordingly, it may be possible to prevent voids from occurring between the first backside insulating layer **26***a* of the first core die **20***a* and the first front insulating layer **22***b* of the second core die **20***b*. For example, when the third carrier substrate C**3** and the second carrier substrate C**2** are bonded to each other by wafer-to-wafer bonding, no bevel step difference is generated between a second surface **14***a* of a silicon substrate **10***a* and a backside surface of the first substrate **21***a* of the first core die **20***a* at an edge region of the third carrier substrate C**3**. Accordingly, it may be possible to prevent a void from occurring between the edge region of the third carrier substrate C**3** and an edge region of the second carrier substrate C**2**.

[0152] Referring to FIG. **26**, a first surface **12***a* of the third carrier substrate C**2** may be partially removed until a front surface of the first core die **20***a* is exposed.

[0153] In example embodiments, the first surface **12***a* of the third carrier substrate C**3** may be partially removed by a grinding process. The first surface **12***a* of the third carrier substrate C**3** may be removed to expose a bottom surface of a recess RC**3**. Accordingly, the recess RC**3** may become an opening **16***a* that penetrates the silicon substrate **10***a*. The first substrate **21***a* and the first front insulating layer **22***a* may be disposed within the opening **16***a* of the silicon substrate **10***a*. The first front insulating layer **22***a* of the first core die **20***a* may be exposed by the opening **16***a*. Accordingly, the first bonding pads **23***a* may be exposed from the first surface **12***a* of the silicon substrate **10***a*.

[0154] The first core die **20***a* may be disposed in the opening **16***a* of the silicon substrate **10***a*, and the gap filling layer **30***a* may be formed to fill a gap between an outer side surface of the first core die **20***a* and a sidewall of the opening **16***a*.

[0155] Referring to FIG. **27**, the first, second and third carrier substrate C**1**, C**2** and C**3** of FIG. **26** (in which the first, second and third core dies **20***a*, **20***b* and **20***c* are stacked) may be attached on a buffer wafer BW (in which a buffer die **60** is formed) by wafer-to-wafer hybrid bonding process.

[0156] In example embodiments, the structure of FIG. **26** may be turned over, and the third carrier substrate C**3** may be bonded on the buffer wafer BW including the buffer dies **60**. A front surface of the first core die **20***a* may face a backside surface of the buffer die **60** of the buffer wafer BW by the bonding.

[0157] When the third carrier substrate C**3** and the buffer wafer BW are bonded to each other by wafer-to-wafer bonding (e.g., by a thermal compression process), the first core die **20***a* of the third carrier substrate C**3** and the buffer die **60** of the buffer wafer BW and may be bonded to each other in a manner of hybrid bonding. For example, the bonding pad **23***a* of the first core die **20***a* and the second bonding pad **67** of the buffer die **60** may be directly bonded to each other by Cu—Cu bonding. Additionally, the backside surface of the buffer die **60** (i.e., a backside insulating layer **66** on a backside surface of a substrate **61**) may be directly bonded to the front surface of the first core die **20***a* (i.e., the first front insulating layer **22***a* on a first surface **212***a* of the first substrate **21***a*). The first bonding pad **23***a* of the first core die **20***a* and the second bonding pad **17** of the buffer die **60** may be bonded to each other by Cu—Cu bonding (e.g., pad to pad direct bonding). Referring to FIG. **28**, the buffer wafer BW, the first to third carrier substrates C**1**, C**2**, C**3** and the second wafer W**2** may be cut along a cutting region CR to form a plurality of sub-packages **500** of FIG. **20**.

[0158] FIG. **29** is a cross-sectional view illustrating a semiconductor package in accordance with example embodiments. The semiconductor package includes the sub-package illustrated in FIGS. **1** and **20**, however the present inventive concept is not limited thereto.

[0159] Referring to FIG. **29**, a semiconductor package **100** may include a package substrate **110**, a first sub-package **200**, and a second sub-package **500**. Additionally, the semiconductor package **100** may further include first and second underfill members **250** and **550** and a sealing member **600**. The first sub-package **200** may be substantially the same as or similar to the sub-package described with reference to FIG. **1**. The second sub-package **500** may be substantially the same as or similar to the sub-package described with reference to FIG. **20**. Thus, same reference numerals will be used to refer to the same or like elements and any further repetitive explanation concerning the same or like elements will be omitted.

[0160] In example embodiments, the semiconductor package **100** may be provided as a portion of a memory module having a 2.5 D package structure. In this case, the package substrate **110** may be or include an interposer for electrically connecting the first and second sub-packages **200** and **500** to each other.

[0161] In example embodiments, the first sub-package **200** may be or include a logic device, and the second sub-package **500** may be or include a memory device. The logic device may be an application specific integrated circuit (ASIC) chip including, at least one of a graphics processing unit (GPU), a central processing unit (CPU), a microprocessor, a microcontroller, an application processor (AP), a digital signal processing core (digital signal processing core), etc. The memory device may include, for example, DRAM, SRAM, flash, PRAM, ReRAM, FeRAM, or MRAM. The memory device may be a HBM device.

[0162] In example embodiments, the package substrate **110** may have an upper surface and a lower surface opposite to the upper surface, and may be, for example, a printed circuit board (PCB), a silicon interposer, or a redistribution interposer. The printed circuit board may be a multilayer circuit board having various circuit patterns therein.

[0163] The first sub-package **200** may be mounted on the package substrate **110**. The first sub-package **200** may be mounted on an upper surface of the package substrate **110** using a flip chip bonding method. The first sub-package **200** may be arranged such that a first front insulating layer **22** faces the package substrate **110**. The first bonding pads **23** of the first sub-package **200** may be

electrically connected to substrate pads of the package substrate **110** through conductive bumps **50**.
[0164] The second sub-package **500** may be horizontally spaced apart from the first sub-package **200** on the package substrate **110**. The second sub-package **500** may be mounted on the upper surface of the package substrate **110** via conductive bumps **70**.
[0165] The first sub-package **200** and the second sub-package **500** may be electrically connected to each other through wires inside the package substrate **110**. The package substrate **110** may provide high-density interconnection between the first and second sub-packages **200** and **500**.
[0166] In example embodiments, the first and second underfill members **250** and **550** may include a material with relatively high fluidity to effectively fill small spaces between the first and second sub-packages **200** and **500** and the package substrate **110**. For example, the first and second underfill members **250** and **550** may include an adhesive containing an epoxy material.
[0167] The sealing member **600** may be provided on the upper surface of the package substrate **110** to cover the first and second sub-packages **200** and **500**. For example, the sealing member **600** may include an epoxy mold compound (EMC). The sealing member **600** may include UV resin, polyurethane resin, silicone resin, silica fillers, etc.
[0168] Although not illustrated in the figures, a heat slug may cover and be in thermal contact with the first and second sub-packages **200** and **500** on the package substrate **110**. In this case, the sealing member **600** may be omitted. Alternatively, the sealing member **600** may expose upper surfaces of the first and second sub-packages **200** and **500**, and a heat dissipation member may be disposed on the upper surfaces of the first and second sub-packages **200** and **500** exposed by the sealing member **600**. The heat dissipation member may include, for example, a thermal interface material (TIM). The heat slug may be in thermal contact with the first and second sub-packages **200** and **500** via the heat dissipation member.
[0169] The semiconductor package **100** may include semiconductor devices such as logic devices or memory devices. The semiconductor package may include logic devices such as central processing units (CPUs), main processing units (MPUs), or application processors (APs), or the like, and volatile memory devices such as DRAM devices, HBM devices, or non-volatile memory devices such as flash memory devices, PRAM devices, MRAM devices, ReRAM devices, or the like.
[0170] The foregoing is illustrative of example embodiments and is not to be construed as limiting thereof. Although a few example embodiments have been described, those skilled in the art will readily appreciate that many modifications are possible in example embodiments without materially departing from the novel teachings and advantages of the present invention. Accordingly, all such modifications are intended to be included within the scope of example embodiments as defined in the claims.

Claims

1. A semiconductor package, comprising: a first die structure including a support substrate having an opening therein, a first semiconductor chip disposed in the opening, and a gap filling layer filling a gap between a sidewall of the opening and the first semiconductor chip; and a second die structure stacked on the first die structure, the second die structure including a second semiconductor chip electrically connected to the first semiconductor chip, wherein the first semiconductor chip includes a first circuit substrate, a first front insulating layer formed on a first surface of the first circuit substrate and provided with first bonding pads, and a first backside insulating layer formed on a second surface of the first circuit substrate and provided with second bonding pads, and the second surface of the first circuit substrate is positioned opposite to the first surface of the first circuit substrate, wherein the second semiconductor chip includes a second circuit substrate, and a second front insulating layer formed on a first surface of the second circuit substrate and provided with third bonding pads, and wherein the second bonding pads are directly bonded to the third bonding pads respectively.

2. The semiconductor package of claim 1, wherein the first backside insulating layer is directly bonded to the second front insulating layer.

3. The semiconductor package of claim 1, wherein the support substrate is a silicon substrate.

4. The semiconductor package of claim 1, wherein the gap filling layer is formed of silicon oxide.

5. The semiconductor package of claim 1, wherein the first backside insulating layer laterally extends along the second surface of the first circuit substrate to cover the gap filling layer.

6. The semiconductor package of claim 1, wherein an outer side surface of the second semiconductor chip and an outer side surface of the support substrate are coplanar.

7. The semiconductor package of claim 1, wherein the second surface of the first circuit substrate of the first semiconductor chip and an upper surface of the gap filling layer are positioned on the same plane.

8. The semiconductor package of claim 1, wherein the second surface of the first circuit substrate of the first semiconductor chip and an upper surface of the support substrate are positioned on the same plane.

9. The semiconductor package of claim 1, wherein the support substrate has a first coefficient of thermal expansion, and the gap filling layer has a second coefficient of thermal expansion that is greater than the first coefficient of thermal expansion.

10. The semiconductor package of claim 1, further comprising: conductive bumps provided on the first bonding pads, respectively.

11. A semiconductor package, comprising: a first die structure including: a support substrate having a first surface and a second surface opposite to the first surface and having an opening that extends from the first surface to the second surface; a first semiconductor chip disposed in the opening; and a gap filling layer that fills a gap between a sidewall of the opening and the first semiconductor chip; and a second die structure stacked on the first die structure, the second die structure including a second semiconductor chip that is electrically connected to the first semiconductor chip, wherein an outer side surface of the support substrate and an outer side surface of the second semiconductor chip are positioned on the same plane.

12. The semiconductor package of claim 11, wherein the first semiconductor chip includes: a first circuit substrate, a plurality of through electrodes penetrating the first circuit substrate, first bonding pads provided on the first surface of the first circuit substrate and electrically connected to the plurality of through electrodes, and second bonding pads provided on a second surface of the first circuit substrate opposite to the first surface and electrically connected to the plurality of through electrodes, wherein the second semiconductor chip includes a second circuit substrate and third bonding pads provided on a first surface of the second circuit substrate, and wherein the first bonding pads are directly bonded to the second bonding pads.

13. The semiconductor package of claim 12, wherein the first semiconductor chip further includes a first backside insulating layer provided on the second surface of the first circuit substrate and exposing at least portions of the second bonding pads, wherein the second semiconductor chip further includes a second front insulating layer provided on the first surface of the second circuit substrate and exposing at least portion of each of the third bonding pads, and wherein the first backside insulating layer is directly bonded to the second front insulating layer.

14. The semiconductor package of claim 13, wherein the first backside insulating layer laterally extends along the second surface of the first circuit substrate to cover the gap filling layer.

15. The semiconductor package of claim 13, wherein the second surface of the first circuit substrate of the first semiconductor chip and an upper surface of the gap filling layer are positioned on the same plane.

16. The semiconductor package of claim 12, wherein the second surface of the first circuit substrate of the first semiconductor chip and an upper surface of the support substrate are positioned on the same plane.

17. The semiconductor package of claim 11, wherein, the support substrate is a silicon substrate.
18. The semiconductor package of claim 11, wherein the gap filling layer is formed of silicon oxide.
19. A semiconductor package, comprising: a first semiconductor chip; a plurality of die structures sequentially stacked on the first semiconductor chip; and a second semiconductor chip stacked on an uppermost die structure of the plurality of die structures, wherein each of the plurality of die structures includes: a support substrate having an opening therein, a third semiconductor chip disposed in the opening, and a gap filling layer that fills a gap between a sidewall of the opening and the third semiconductor chip, and wherein an outer side surface of the support substrate of the uppermost die structure and an outer side surface of the second semiconductor chip are positioned on the same plane.
20. The semiconductor package of claim **21**, wherein the first semiconductor chip is a semiconductor buffer chip, and the second and third semiconductor chips are semiconductor memory chips.
